

Compliance Demand May Drive Carbon Prices to Rebound

履约需求或使碳价筑底反弹

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Abstract 摘要

1. Calculation of Carbon Allowance Surplus or Deficit in the National Carbon Market Over the Years

A. The net increase in carbon allowances for the first three compliance periods of the national carbon market were 317mn t, 14mn t, and -29mn t, respectively; taking into account the usage of CCERs and the compliance completion rate of key emitting entities during the first three compliance periods; the total surplus of carbon allowances in the market was approximately 368mn t at the beginning of 2024.

B. Assuming the company's carbon emission intensity decreased by 0.5% in 2024 compared to 2023, and considering the introduction of an appropriate correction factor for the average carbon emission intensity of the Utilities & EP sector based on the statistics bureau's published data on thermal power generation above the designated size, the estimated net increase in market carbon allowances for 2024 was -29mn t; the total surplus of market carbon allowances at the beginning of 2025 was approximately 345mn t.

C. Under the scenario of full market quota carry-over, quotas are either used for compliance or carried forward to the next year if they meet the requirements. When enterprises with surplus quotas reasonably plan their quota transactions to meet the carry-over requirements, the banking policy will not result in market quotas "disappearing out of thin air"; the total surplus of market carbon quotas at the beginning of 2026 will be approximately 316mn t.

2. The national carbon market may rebound from its bottom.

In Oct, the national carbon market trading volume was 41.5625mt, an increase of 27.1% compared to Sep, further reducing the passive quota sales volume. Calculations show that under three trading model assumptions, if we do not consider the proactive over-selling by enterprises, the passive quota sales volume has reached 0, indicating that the remaining allowances of quota-surplus enterprises may have already met the maximum carry-over requirement, leading to a decrease in the willingness to sell and confirming the bottom support for carbon prices. **As compliance demand continues to release in Nov, carbon prices may sustain a rebound until the compliance demand subsides. It is necessary to monitor daily trading volume data; a reduction in trading volume could signal a risk of a pullback in carbon prices after the rebound.**

Risks: 1) Limitations of trading model assumptions; 2) Data precision constraints in the calculation of quota surpluses and deficits; 3) Diversity in corporate trading behaviors; 4) Carbon market policies.

1. 历年碳配额盈缺测算

结论一：全国碳市场前三个履约周期市场碳配额净增量分别为 3.17 亿吨、0.14 亿吨及-0.29 亿吨；综合考虑前三个履约周期中 CCER 使用量及重点排放企业的履约完成率等因素，2024 年初市场碳配额总盈余约 3.68 亿吨。

结论二：假设企业碳排放强度 2024 年较 2023 年下降 0.5%，根据统计局公布的规模以上火力发电量，并对电力行业平均碳排放强度考虑引入适当的修正系数进行修正，测算得 2024 年度市场碳配额净增量为-0.29 亿吨；2025 年初市场碳配额总盈余约 3.45 亿吨。

结论三：市场配额完全结转情景下，配额或用于履约，或满足结转条件结转至明年使用。在配额盈余企业合理规划配额交易以满足结转要求的情况下，结转政策并不会导致市场配额“凭空消失”；2026 年初市场碳配额总盈余约 3.16 亿吨。

2. 全国碳市场筑底反弹

10 月全国碳市场成交量为 4156.25 万吨，交易量较 9 月份增长了 27.1%，被动卖出的配额余额进一步减小。经测算，在三种交易模型假设下，若不考虑企业的主动超额卖出情况，被动卖出的配额余额已等于 0，表明配额盈余企业剩余配额或已满足最大可结转量的要求，企业卖出意愿降低，碳价的底部支撑已得到确认。**随着 11 月履约需求的进一步释放，碳价或将持续反弹至履约需求退场。需关注日成交量数据，成交缩量则需警惕碳价反弹后的回调风险。**

风险因子：1) 交易模型假设的局限性；2) 配额盈缺测算过程数据精度受限；3) 企业交易策略、交易行为的多样性；4) 碳市场政策。

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1. Research Background and Assumptions

1.1 Background

The carbon price has been declining throughout 2025 under the "dominance" of the allowance banking policy. After reaching a historical high of Rmb106.02/t on Nov 13, 2024, the carbon price entered a downward trend and continued to fall. Despite a cumulative trading volume increase of 150.3% YoY in the first three quarters of 2025, the carbon allowance closing price dropped by 40.5% compared to the beginning of the year.

The research shows that both buyers and sellers in the market are largely in a "double blind" state. Many key emitting entities lack a clear understanding of the potential sell-offs due to surplus and banking policies, as well as the compliance demand for quotas. For most participating companies, the buying and selling of quotas in the market is in a "double blind" state. Moreover, since the beginning of the year, the carbon market has experienced several instances of rapid quota price declines, which have further intensified market panic.

1.2 Research purpose

This report, based on comprehensive market information and interviews with industry participants, integrates the current status of the Utilities & EP sector, proposes several assumptions, and conducts calculations on various aspects such as the carbon market quota surplus, enterprise's quota surplus volume, quota deficit, enterprise's quota deficit volume, and the market's net quota surplus volume. It further discusses the possibilities of different trading behaviors and evaluates the impact of the banking policy under various trading models.

1.3 Assumptions

(1) In quota trading, the transaction volumes each month and quarter that are driven by compliance requirements do not remain entirely consistent. To simplify the trading model, this article assumes that the proportion of transactions throughout the year is consistent, without additional assumptions.

(2) In fact, with the significant improvement in carbon emission management capabilities of key emission entities, the number of enterprises with quota shortages has been decreasing. However, both the carbon emission balance and benchmark values have also been declining annually. There is no complete data to support the difference in the rate of decline between the two. Here, we assume that they decrease synchronously and proportionally, meaning that the quota shortage rate remains at a low level and constant across each compliance period.

(3) The diversity of trading strategies and behaviors cannot be fully enumerated. In this report, when estimating the impact of the quota banking policy, it is assumed that most quota surplus enterprises adopt a strategy to ensure the maximum quota banking volume, which means selling 40% of their surplus quotas and carrying over the remaining quotas for use in subsequent years' compliance. This assumption does not fully cover various trading strategies such as selling first and then buying, so the results of the quota balance sheet have certain limitations. To address these limitations, Section (2) of Part Four of the report includes a discussion on how changes in the "proactive excess quota selling" ratio affect the "passive quota sales volume".

Chart 1.CEA prices and traded volumes



Source: Wind, CITIC Futures

2. Carbon Market Allowance Surplus Calculation

2.1 The carbon allowance surplus and deficit in the first three compliance cycles

To gain a clearer understanding of the current market supply and demand situation for carbon allowances, it is first necessary to calculate the surplus or deficit of carbon allowances for each compliance period.

The net surplus of carbon allowances for the first compliance period (2019-2020) was approximately 317mn t. The "National Carbon Emission Trading Market First Compliance Period Report (2022)" released by the Ministry of Ecology and Environment did not mention the allocation and surrender volumes of allowances during the first compliance period (2019-2020). It only pointed out that the first compliance period included 2,162 key emitting entities, covering an annual CO₂ emission volume of about 4.5bn t, among which 847 key emitting entities had a shortfall in allowances, totaling approximately 188mn t (accounting for about 2.1% of the total surrender volume). According to Professor Lü Xuedu, the chief climate change expert at the Asian Development Bank, including the nationally certified voluntary

emission reductions (CCERs) available for trading and the allocated allowances, the national carbon market had a surplus of about 360mn t in 2021. After deducting the CCERs available for trading, which amounted to about 42.92mn t, the actual net surplus of carbon allowances for the first compliance period was approximately 317mn t (net surplus refers to the surplus allowances of key emitting entities with excess allowances minus the shortfall allowances of entities with a deficit; a positive value indicates an overall market surplus, while a negative value indicates an overall market deficit). Based on this, it can be further calculated that the total surplus of allowances for entities with excess allowances during the first compliance period was about 505mn t.

The net surplus of carbon allowances for the second and third compliance periods (2021, 2022, and 2023) were 0.01mn t, 1.30mn t, and -2.90mn t, respectively. According to the "National Carbon Market Development Report (2024)" and the "National Carbon Market Development Report (2025)" released by the Ministry of Ecology and Environment, the net surplus of carbon allowances in the national carbon emissions trading market for 2021, 2022, and 2023 were 1.47mn t, 12.98mn t, and -29mn t, respectively (a total deficit of -29mn t indicates a shortfall of 29mn t in allowances). For the second and third compliance periods, the Ministry of Ecology and Environment did not formally release the annual shortfall volumes of enterprises with allowance deficits. To obtain the data on allowance surpluses and deficits, we assume that the proportion of enterprises with allowance deficits in the power generation sector during the second and third compliance periods is the same as in the first compliance period. Based on this assumption, the total annual allowance deficits for the second and third compliance periods (2021, 2022, and 2023) can be estimated at 107mn t, 107mn t, and 110mn t, respectively. Consequently, the gross annual allowance surpluses for enterprises with surpluses can be calculated at 108mn t, 120mn t, and 81mn t, respectively.

Chart 2: Comparison of the Advantages and Disadvantages of Different Solid-State Electrolytes

Years	Period	Surplus of allowances 10kt	Deficit of allowances 10kt	Net surplus of allowances 10kt	Percentage of net surplus in total demands %
2019-2020	First	50500	-18800	31700	3.37%
2021		10844	-10697	147	0.03%
2022	Second	11989	-10691	1298	0.25%
2023	Third	8112	-11012	-2900	-0.55%

Source: The First Compliance Period Report of the National Carbon Emission Trading Market (2022), The Development Report of the National Carbon Market (2024), The Development Report of the National Carbon Market (2025), CITIC Futures

2.2 The carbon allowance surplus and deficit in the fourth compliance cycle

The calculation of the surplus or deficit of allowances for the fourth compliance period does not need to consider the impact of the three newly added sectors. The Ministry of Ecology and Environment has not yet announced the total amount of allowances issued and the amount of allowances due for surrender for the fourth compliance period. Moreover, during this period, the steel, cement, and electrolytic aluminum sectors were newly included, adding 1,334 key emission entities. According to the "Draft Allocation Plan for Total Allowances and Distribution for the Steel, Cement, and Aluminum Smelting Sectors in the National Carbon Emission Trading Market for the Years 2024 and 2025" released by the Ministry of Ecology and Environment on Sep 30, 2025, the allowance allocation methods for the steel, cement, and electrolytic aluminum sectors are generally consistent with those for the power generation sector. The allowances for the years 2024 and 2025 are allocated free of charge, based on the carbon emissions per unit of output. Among these, 2024 was the first compliance year for the three sectors, and the allowances received by each key emission entity in these sectors were equal to their verified actual carbon emissions, meaning there was no allowance shortfall and no substantial compliance pressure. Therefore, when estimating the surplus or deficit of allowances for the fourth compliance period, the impact of the three newly added sectors is not considered.

(1) The total allocation of allowances for the fourth compliance period.

According to the guidance requirements of the "National Carbon Emission Allowance Allocation Plan for the Power Generation Sector for the Years 2023 and 2024", all allowances for the years 2023 and 2024 are allocated free of charge, using the benchmark method combined with an exemption mechanism at the unit level to determine the allowance allocation for each unit. The annual allowance allocation for each key emission entity is calculated by summing up the annual allowance allocations for all units owned by the entity and then applying the exemption mechanism at the key emission entity level. The total allowance is proportional to the thermal power generation volume of regulated enterprises in the power generation industry. However, due to the need to comprehensively consider factors such as compliance preferential policies, peak load adjustment correction coefficients, and the 20% upper limit exemption mechanism for allowance compliance gaps, as well as the varying data collection methods for the power generation volumes of different units, the annual allowance allocation and the total amount of allowances to be surrendered cannot be directly calculated. Therefore, based on the carbon emission balance values and benchmark values published by the Ministry of Ecology and Environment for each year, this report introduces the concepts of "comprehensive carbon emission balance value" and "comprehensive carbon emission benchmark value correction coefficient" to simplify the calculation of allowance allocation and

the amount of allowances to be surrendered. Specifically, the calculation is conducted based on the thermal power generation volume above designated scale, using the following formula:

Annual allowance surrender = Annual carbon emission intensity balance value × Comprehensive carbon emission balance coefficient α1 × Annual thermal power generation

Annual carbon emission allowance allocation = annual carbon emission intensity benchmark × comprehensive carbon emission benchmark correction factor α2 × annual thermal power generation

The comprehensive carbon emission balance adjustment coefficient α1 — taking into account the power generation proportion of conventional coal-fired units above 300MW, conventional coal-fired units 300MW and below, unconventional coal-fired units such as coal gangue, coal sludge, and water-coal slurry (including circulating fluidized bed coal-fired units), and gas-fired units, this coefficient is used to weight and adjust the carbon emission balance values of each unit, thereby directly calculating the required quota surrender.

The comprehensive carbon emission benchmark adjustment factor α2 — taking into account the power generation ratio of conventional coal-fired units above 300MW, conventional coal-fired units at or below 300MW, unconventional coal-fired units such as coal gangue, coal slurry, and water-coal slurry units (including circulating fluidized bed coal-fired units), and gas-fired units, this factor is used to weight and adjust the carbon emission benchmarks of each unit type, and is directly applied to calculate the allocation of quotas.

Based on the announced quota allocation and the required quota surrender for the year 2023, the calculated comprehensive carbon emission balance value and the revised comprehensive carbon emission benchmark coefficient were 1.0344 and 1.0355, respectively.

The "National Carbon Emission Allowance Trading Scheme for the Power Generation Sector in 2023 and 2024" provides the carbon emission balance values and benchmark values for 2023, as well as the carbon emission benchmark values for 2024, as shown in the table below: The carbon emission benchmark value for 2024 decreased by 0.5ppts compared to 2023.

Chart 3: Benchmark Values of Generator/Heating Units in 2023 and 2024

Unit classification	Benchmark value of generation unit tCO2/MWh			Benchmark value of heating unit tCO2/GJ		
	Balance value in 2023	Benchmark value in 2023	Benchmark value in 2024	Balance value in 2023	Benchmark value in 2023	Benchmark value in 2024

Conventional coal-fired units above 300MW rating	0.7982	0.7950	0.7910			
Conventional coal-fired units below 300MW rating	0.8155	0.8090	0.8049			
				0.1041	0.1038	0.1033
Unconventional coal-fired units (e.g., burning coal gangue, coal slime, coal water slurry), including coal circulating fluidized bed (CFB) units	0.8352	0.8285	0.8244			
Gas-fired units	0.3239	0.3305	0.3288	0.0525	0.0536	0.0533

Source: National Carbon Emission Allowance Trading Scheme for the Power Sector - Total Allowance and Allocation Plan for 2023 and 2024, CITIC Futures Research Institute, CITIC Futures

Assuming that the comprehensive carbon emission balance value and the comprehensive carbon emission benchmark correction coefficient for 2024 are equal to those for 2023, then based on the power generation volume from thermal power plants above the designated size in 2024, the calculated allocation of allowances and the amount of allowances to be surrendered for the fourth compliance period are:

a) Assumption one: If the carbon emission balance value in 2024 is equal to that in 2023, then the allocation volume for the fourth compliance period would be 5.279bn t, the volume of quotas to be surrendered would be 5.335bn t, and the annual net surplus of quotas would be -56.03mn t;

b) Assumption two: If the carbon emission balance value in 2024 decreases by 0.5% compared to 2023, then the allowance allocation for the fourth compliance period would be 5.279bn t, the allowances to be surrendered would be 5.308bn t, and the annual net surplus of allowances would be -29.35mn t;

Considering that since China proposed the dual carbon targets, the carbon emission intensity of the thermal power generation sector has been declining annually, it is advisable to appropriately reduce the carbon emission intensity balance value when calculating the Utilities & EP sector's required quota surrender for the fourth compliance period. This report adopts the calculation results under assumption two, which indicates that the Utilities & EP sector's quota allocation for the fourth compliance period is 5.279bn t, the required quota surrender is 5.308bn t, and the annual net surplus of quotas is -29.35mn t.

(2) Quota Surplus and Deficit Calculation for the Fourth Compliance Period

To calculate the total surplus and total deficit of enterprises in the fourth compliance period, we assume that the deficit rate of enterprises in the power generation sector during the fourth compliance period is the same as in the previous three compliance periods, i.e., 2.1% of the total quotas required for surrender. Under this assumption, the annual total deficit for enterprises with a quota shortfall in the fourth compliance period is 111mt, and the annual total surplus for enterprises with a quota excess is 82mt.

3. Impact of the Quota Banking Policy

3.1 Quota banking policy requirements

The formula for calculating the maximum carry-over volume of carbon emission quotas for key emitting entities:

$$T = B + N_S \times R \quad (\text{Formula 1})$$

In the formula:

T - The maximum allowable carry-over volume for key emission entities, unit: tCO₂;

B—Basic allowable carry-over volume for key emission entities, Unit: tCO₂;

NS—Net Quota Sales Volume, equal to the Quota Purchase Volume minus the Quota Sales Volume, unit: tCO₂;

R— Carry-over Ratio, the carry-over ratio is set at 1.5;

Other requirements:

(1) The allowable carry-over volume shall not exceed the key emission unit's holding volume at the end of 2025; the basic allowable volume for key emission units is 10,000t; the net sales volume calculation period: from Jan 1, 2024, to Dec 31, 2025, if the result is less than 0, it shall be taken as 0; the maximum allowable volume for key emission units shall be rounded down to the nearest whole number; the holding volume at the end of the period is the allowance holding volume shown in the National Carbon Emission Rights Registration System at 23:59 on Dec 31, 2025, for the 2019-2024 allocation period.

(2) The national carbon emission allowance registration institution will conduct the quota recovery and carry-over process in two phases: the first phase will handle carry-over applications submitted up to Mar 20, 2026, before Mar 27, 2026; the second phase will handle applications submitted from Mar 21, 2026, to Jun 10, 2026, before Jun 26, 2026.

According to [Formula 1](#), the maximum allowable amount for key emission entities is proportional to the net sales volume during the period from Jan 1, 2024, to Dec 31, 2025, and all enterprises are entitled to a basic allowable amount of 10kt.

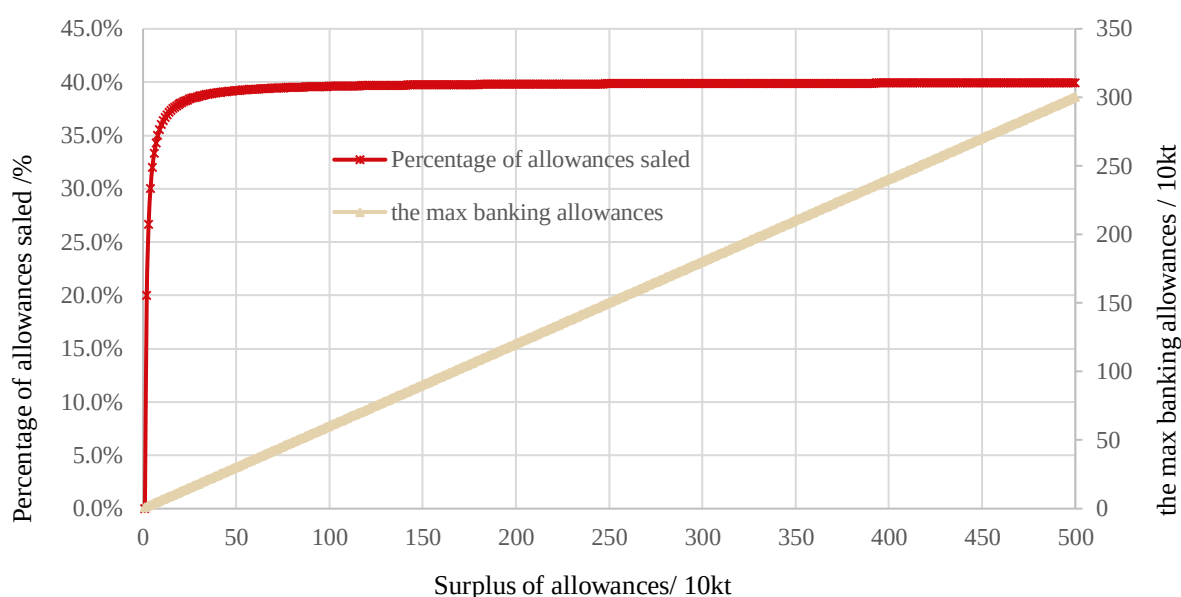
3.2 Maximum Carry-over Trading Strategy

(1) Quota-deficit enterprises: Most quota-deficit enterprises often have no excess carbon quotas in their accounts and must purchase quotas from the market to ensure full compliance. The net sales volume for such enterprises is typically negative, hence the maximum carry-over amount is 10kt.

(2) Quota-surplus enterprises: Quota-surplus enterprises need to determine the surplus quotas in their accounts (after deducting the annual quota surrender requirement) according to Formula 1. A portion of the surplus quotas must be sold, with the remaining quotas carried over. Otherwise, after the transaction deadline specified by the banking policy, any quotas exceeding the maximum allowable carry-forward amount in the enterprise's carbon quota account will be invalidated. Therefore, quota-surplus enterprises need to plan the carry-over amount and the quota amount that must be sold (hereinafter referred to as "passive quota sales volume") to achieve the allowable carry-over amount.

If quota-surplus enterprises "hoard", they may tend to adopt a trading strategy that ensures the maximum allowable carry-over volume of surplus quotas (hereinafter referred to as the "maximum carry-over trading strategy"). Under this strategy, enterprises passively sell quotas and plan to carry over all unsold remaining quotas to meet future annual compliance requirements. The relationship between the passive sales volume, the passive sales ratio, and the account quota volume for enterprises adopting this trading strategy is shown in the following figure.

Chart 4: Relationship between the amount of passive sales/the proportion of passive sales and the amount quota volume



Source: CITIC Futures

From Chart 4, it can be seen that if a company adopts the "maximum carry-over trading strategy", then for companies with a larger surplus of quotas, such as those exceeding 100mt, the carry-over quota volume and the passive sales volume would be 60.4mt and 39.6mt, respectively. This means that companies are required to sell approximately 40% of their surplus quotas during the period from Jan 1, 2024 to Dec 31, 2025, to ensure that the remaining quotas can be successfully carried over by 2026. For companies with smaller surplus quotas, the proportion of passively sold quotas is less than 40%. For example, companies with surplus quotas of 10mt, 5mt, and 1mt would have their passive quota sales proportions at 36.0%, 32.0%, and 0.0%, respectively. In summary, if all quota-surplus companies adopt the "maximum carry-over trading strategy", the proportion of all passively sold quotas by these companies would be less than 40% of the total surplus quotas of all quota-surplus companies in the market.

However, it should also be noted that different companies have varying views on the trends in carbon allowance prices, national carbon market policies, and their past participation in trading, which can directly or indirectly influence them to adopt different allowance trading strategies. Companies with surplus allowances do not all choose the "maximum carry-over trading strategy," and some may "actively over-sell" their surplus allowances, meaning the proportion of surplus allowances sold exceeds 40%, or they might sell all of them. In such a scenario, the total volume of allowances sold "actively" and "passively" by companies with surplus allowances could exceed 40%.

4. Outlook on the National Carbon Market

4.1 Calculation of the "passive quota sales volume"

To calculate the remaining market passive quota sales volume, a market quota balance sheet needs to be constructed.

At the beginning of 2024, the national carbon market had a net surplus of carbon allowances of about 368mn t. Assuming that by the end of 2023, enterprises that did not fully comply during the first compliance period achieved 100% compliance through self-purchase supplements, withholding and payment by the competent authority, and other methods, then based on the previous estimates of the surplus and deficit of allowances for the first and second compliance periods, and taking into account the use of CCERs to offset part of the allowances during these periods, it can be calculated that the national carbon market had a net surplus of about 368mn t at the beginning of 2024. Among these, the total surplus for allowance-surplus enterprises was approximately 394mn t, while the total deficit for allowance-deficit enterprises was 26mn t (the allowance deficit generated by enterprises that did not complete compliance during the second compliance period).

For the full year of 2024, the national carbon market experienced an annual carbon allowance shortfall of about 29mn tonnes. The new annual allowances allocated in 2024 were less than the allowances required for surrender by key emitting entities, resulting in the first-ever annual allowance shortfall, estimated at 29mn tonnes. According to the calculations in Table 2, enterprises with surplus allowances added approximately 81mn tonnes of surplus in 2024, while those with a shortfall added about 110mn tonnes of deficit.

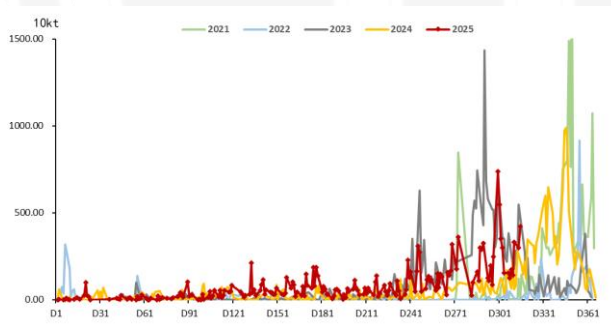
At the beginning of 2025, the national carbon market had a net surplus of carbon allowances of about 345mn t. Assuming that by the end of 2024, all enterprises that did not fulfill their obligations in the second compliance period completed 100% compliance, and taking into account the use of CCERs to offset some allowances in the third compliance period, it can be estimated that at the beginning of 2025, the national carbon market had a net surplus of carbon allowances of about 345mn t, where the total surplus of allowance-surplus enterprises was about 346mn t, and the shortfall of allowance-deficit enterprises was 1mn t (the allowance shortfall generated by enterprises that did not complete their obligations in the third compliance period).

For the whole year of 2025, the national carbon market is expected to have an annual carbon allowance shortfall of about 29mn t (the allocation of carbon allowances for the utilities & ep sector in the fourth compliance period was completed by Aug 20, 2025). The fourth compliance period have included three new major sectors: steel, cement, and aluminum smelting. However, according to the "Draft Plan for the Total Allowance and Allocation Scheme for Steel, Cement, and Aluminum Smelting Sectors in the National Carbon Emission Trading Market for 2024 and 2025", the carbon allowances allocated to key emission units in the steel, cement, and aluminum smelting sectors for 2024 were equal to their verified actual carbon emissions, resulting in a net allowance surplus of 0 for these three sectors, thus having no impact on the overall allowance surplus of the national carbon market. However, if the 1334 newly added controlled emission enterprises purchase carbon allowances based on the basic carry-over volume and apply for carry-over in 2026, it will create a procurement demand of 13.34mn t for the market.

As of Sep 30, 2025, the total quota trading volume in the national carbon market was 98.0988mn t. To calculate the net sales volume, it is necessary to determine the proportion of purchasing demand from enterprises with a quota shortfall in the total quota trading volume of the national carbon market. The actual quota purchases made by enterprises with a shortfall for compliance purposes in 2024 and 2023 were equal to the current year's quota shortfall + the previous year's unfulfilled quota - CCER offset volume, resulting in the proportion of quotas purchased by shortfall enterprises for compliance purposes being 68.2% and 70.1% of the annual quota trading volume, respectively. Assuming that the trading behavior of most market participants in 2025 did not change significantly and that the proportion of quota

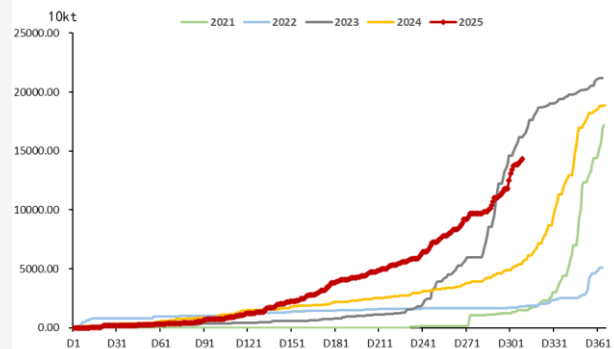
transactions driven by compliance needs remained consistent across all four quarters, the quota trading volume driven by compliance procurement in the first three quarters of 2025 would be approximately 68.6692mn t, accounting for about 70% of the total quota trading volume in the first three quarters (Assumption One of the Trading Model). However, 2025 is the final year of the quota banking policy, and the quota trading volume in the first three quarters of 2025 increased by 150.3% YoY. If we assume that this growth was entirely due to purchases by enterprises with a quota shortfall (as enterprises with a surplus quota need to sell quotas to complete the carryover), then the quota trading volume driven by compliance procurement in the first three quarters of 2025 would be approximately 86.3396mn t, accounting for about 88% of the total quota trading volume in the first three quarters (Assumption Two of the Trading Model). According to past quota trading patterns, the release of compliance procurement demand primarily concentrates in the third and fourth quarters of each year. Under Assumption Two, considering the actual trading situation, the reporting and pre-allocation time for power sector emissions, the proportion of compliance demand in the first quarter's quota trading is relatively low. After adjustment, the quota trading volume driven by compliance procurement in the first three quarters of 2025 is approximately 81.815mn t, accounting for about 83% of the total quota trading volume in the first three quarters (Assumption Three of the Trading Model).

Chart 5: 2021-2025 CEA Daily Trading Volume Comparison



Source: Wind, CITIC Futures

Chart 6: 2021-2025 CEA Cumulative Transaction Volumes Comparison



Source: Wind, CITIC Futures

Calculate the market's passive quota sales volume for the fourth quarter of 2025 under three different trading model assumptions:

When estimating the total quota trading volume driven by compliance procurement needs for the full year of 2025, assuming all key emission entities complete their compliance obligations by Dec 31, 2025, i.e., a compliance rate of 100%; then for the full year of 2025, the compliance gap for quota-deficit enterprises is approximately 112mn t.

(1) **Trading Model Assumption One:** The quota trading volume dominated by compliance procurement demand is 68.6692mt, accounting for about 70% of the total quota trading volume in the first three quarters of the year. Under this assumption, taking into consideration factors such as the offset volume of CCERs and the completion of compliance, if quota-surplus enterprises adopt a "hoarding" trading strategy aimed at maximizing the allowable carry-over volume, then by Sep 30, the market's passive quota sales volume is approximately 23.6562mt, and the procurement demand balance is approximately 43.8508mt.

(2) **Trading Model Assumption Two:** The quota trading volume dominated by compliance procurement demand is approximately 86.3396mt, accounting for about 88% of the total quota trading volume in the first three quarters. Under this assumption, taking into consideration factors such as the offset volume of CCERs and the completion of compliance obligations, if all quota-surplus enterprises adopt a "hoarding" trading strategy to ensure the maximum allowable carry-over volume, then by Sep 30, the market's passive quota sales volume is about 5.9984mt, and the procurement demand balance is about 26.1931mt.

(3) **Trading Model Assumption Three:** The quota trading volume dominated by compliance procurement demand is approximately 81.815mt, accounting for about 83% of the total quota trading volume in the first three quarters of 2025. Under this assumption, taking into consideration factors such as the offset volume of CCERs and the completion of compliance, if all quota-surplus enterprises adopt a "hoarding" trading strategy to ensure the maximum allowable carry-over volume, then by Sep 30, 2025, the market's passive quota sales volume would be around 10.9033mt, and the procurement demand balance would be around 31.098mt.

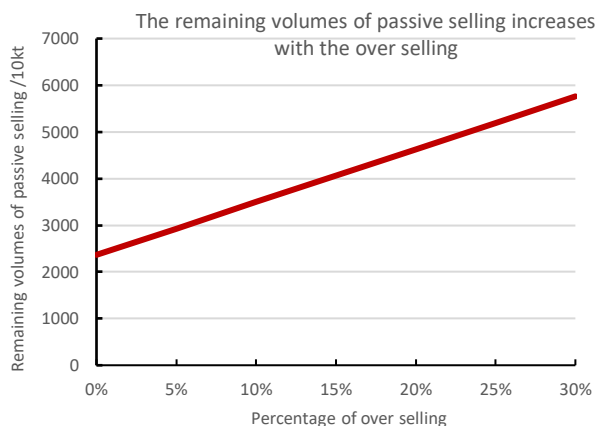
4.2 Impact of the active excess sales volume

As mentioned above, not all quota-surplus companies choose the "maximum carry-over trading strategy", or they may "actively oversell" their surplus quotas, meaning the proportion of surplus quotas sold exceeds 40%, and might even be entirely sold. In this case, the total amount of quotas sold by quota-surplus companies, both "actively" and "passively", will exceed 40%. The larger the proportion of surplus quotas that quota-surplus companies "actively oversell", the more the balance of quotas passively sold in the fourth quarter, and the greater the market sell-off pressure. Under the three trading model assumptions, the balance of quotas passively sold in 4Q25 changes with the proportion of surplus quotas actively oversold by quota-surplus companies as shown in the following chart.

Considering the limitations of the three assumptions, Chart 10 further illustrates the trend of the remaining passive quota sales volume and the compliance purchase balance under Assumption One of the trading model, with changes in the proportion of compliance transactions in the first three quarters of

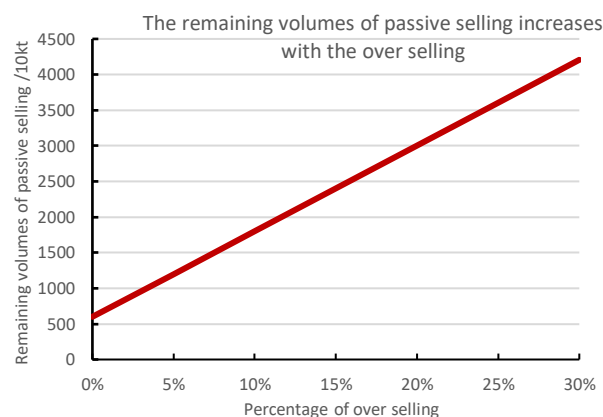
2025: the higher the proportion of compliance transactions, the fewer the remaining passive quota sales volume, and the fewer the remaining compliance purchases.

Chart 7: The relationship between the remaining passive quota sales volume and the "voluntary over-selling" ratio (Assumption One)



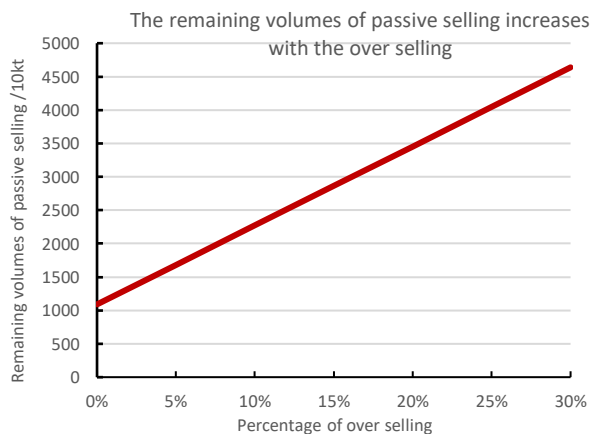
Source: CITIC Futures

Chart 8: The relationship between the remaining passive quota sales volume and the "voluntary over-selling" ratio (Assumption Two)



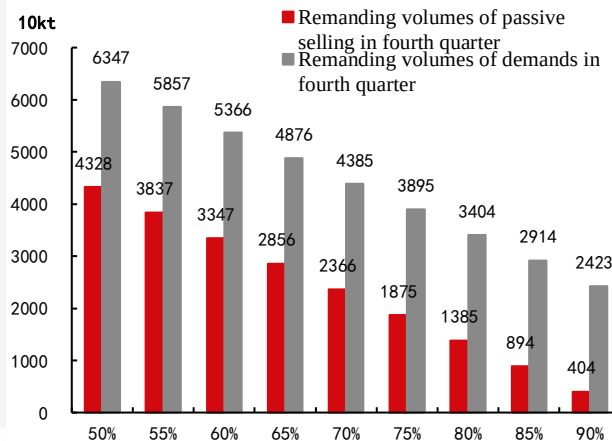
Source: CITIC Futures

Chart 9: The relationship between the passive quota sales volume and the "voluntary over-selling" ratio (Assumption Three)



Source: CITIC Futures

Chart 10: Passive Quota Sales Volume and Compliance Procurement Balance (Assumption One)



Source: CITIC Futures

4.3 Carbon prices may show stronger volatility

Enterprises may be less willing to sell allowances. In Oct, the national carbon market trading volume was 41.5625mt, an increase of 27.1% compared to Sep. The passive quota sales volume further

decreased. According to calculations, under three trading model assumptions, if we do not consider the proactive quota over-selling by enterprises, the passive quota sales volumes has reached 0, indicating that the remaining allowances of surplus enterprises may have met the maximum allowable carry-over amount requirement. Enterprises, considering the "reluctance to sell," may reduce their willingness to sell allowances. Similarly, as enterprises with allowance deficits continue to purchase, the demand for compliance allowances has further decreased (some enterprises have yet to complete their compliance purchases, i.e., demand > 0). Apart from compliance needs, some enterprises might consider using the basic carry-over policy to purchase allowances at lower carbon prices and then carry them over (enterprises may adopt this strategy if they believe the current carbon price is undervalued), with this potential procurement demand estimated at around 20mt.

Therefore, the bottom support for carbon prices may have been confirmed. As the compliance demand further releases in Nov, carbon prices may continue to rebound until the compliance demand subsides (around early Dec). During this period, the rapid rise in carbon prices might lead to a need for adjustments, but in the short term, the adjustment space is limited. It is necessary to continuously monitor the daily trading volume of allowances. If there is a continuous decline in volume, it indicates that the compliance demand has been fully released, and allowance trading will enter a low season, posing a risk of price pullback.

In the long term, it is expected that the carbon market will return to a supply-tight state by 2026, with carbon prices potentially rising in a volatile manner. On the supply side, by the beginning of 2026, the net surplus of allowances in the national carbon market will further decrease, and allowance trading may once again shift to a "reluctance to sell" logic, leading the market back to a supply-tight condition. On the demand side, in 2026, the new allowance shortfall from the three additional sectors will bring new demand growth to the market.

Strategy Recommendations: (1) Companies with a quota shortfall can buy quotas at low prices to complete their annual compliance; (2) Companies with a surplus of quotas should confirm whether the remaining quotas meet the maximum allowable carry-over requirements according to the banking policy, and sell the excess part at high prices if applicable; (3) Key emission units from the three newly included sectors in the national carbon market enjoy a basic carry-over volume of 10kt, and can look for opportunities to buy quotas at low prices in the carbon market.

However, it should be noted that the diversity of quota trading strategies among key emission entities may cause the three trading model scenarios to deviate from actual trading conditions. Furthermore, the proportion of market compliance demand in the total quota trading volume and its distribution across different months will affect the accuracy of this report's calculations of companies' net sales volumes. The

actual conversion of potential demand cannot be determined. A small number of companies might, due to decision-making errors or other reasons, engage in a frenzied sell-off of excess quotas in late Dec, leading to significant carbon price fluctuations. This is an unpredictable factor, so companies should try to avoid trading during periods when carbon prices may experience substantial volatility at the end of the year.

Appendix: Chinese Version

1. 研究背景及限定假设

1.1 背景介绍

2025 年碳价在配额结转政策的“支配”下一路走低。2024 年 11 月 13 日碳价达到 106.02 元/吨的历史高点，之后便进入下跌通道。2025 年前三个季度在累计交易量同比增长 150.3%，但碳配额收盘价较年初下跌 40.5%。

调研发现，碳市场买卖双方多处于“双盲”状态。经团队调研发现，很多排放单位对市场碳配额盈余以及结转政策可能带来的配额抛售情况、配额履约需求情况并没有直观判断。对于多数参与企业而言，市场配额买卖处于“双盲”状态。而今年碳市场经历多次配额急速下跌的行情，加剧了市场恐慌。

1.2 研究目的

本篇报告在公开信息、走访调研的基础上，结合电力行业现状，提出多个假设，针对碳市场配额盈余企业配额盈余量、配额缺口企业配额缺口量、市场配额净盈余量等进行测算，并进一步讨论可能的交易行为可，进而评估各种交易模型下，结转政策的影响。

1.3 假设前提

(1) 配额交易中，每个月份、季度的交易量中，由履约需求主导的配额采购交易量并不完全相同，本文为简化交易模型，在无其他补充假设情况下，假设交易占比全年一致；

(2) 实际上，随着重点排放单位碳排放管控能力的显著提升，配额缺口企业数量在减少，然而碳排放平衡值和基准值也在逐年下降，二者之间下降幅度之差并无完整统计数据支撑。在此假设两者同步同幅度下降，即各履约周期内配额缺口率保持在一个较低水平且不变；

(3) 交易策略和交易行为的多样性不能穷举。本篇报告在进行配额结转政策影响的测算时，假设大部分配额盈余企业采用保障最大可结转配额量的策略，即卖出企业盈余配额的 40%，剩余的配额进行结转，以用于后续年度的履约使用。该假设并不能完全覆盖企业先卖后买等多种交易策略，配额平衡表的测算结果具有一定的局限性。为弥补该局限性，报告第四部分第（二）小节加入了“主动超额卖出配额”比例的变化对“被动卖出配额量”影响的讨论。

图表1：全国碳市场碳价走势



资料来源：Wind 中信期货研究所

2. 碳市场配额盈余测算

2.1 第一~第三履约周期碳配额盈缺情况

为了对当前市场碳配额供需形势有更清晰认知，首先要测算每个履约周期碳配额盈缺情况。

第一履约周期（2019-2020 年度）碳配额净盈余约 3.17 亿吨。生态环境部发布的《全国碳排放权交易市场第一个履约周期报告》（2022 年）中并未提及第一个履约周期（2019-2020 年度）内的配额发放量及应清缴配额量，仅指出第一个履约周期共纳入 2162 家重点排放单位，年度覆盖二氧化碳排放量约 45 亿吨，其中有 847 家重点排放单位存在配额缺口，缺口总量约为 1.88 亿吨（约占应清缴配额量的 2.1%）。根据亚洲开发银行首席气候变化专家吕学都教授的测算，包括可用于交易的国家核证的自愿减排量（CCER）和所分配的配额，2021 年年度全国碳市场有盈余约 3.6 亿吨，扣除可用于交易的 CCER 约 4292 万吨，实际第一履约周期内全国碳市场配额净盈余（配额净盈余指全国重点排放单位中配额盈余企业盈余配额扣减配额缺口企业缺口配额，大于 0 则表示市场总体配额盈余，小于 0 则表示市场总体配额缺口）约 3.17 亿吨。据此可进一步测算得第一履约周期内配额盈余企业配额总盈余约 5.05 亿吨。

第二、三履约周期（2021、2022、2023 年度）碳配额净盈余分别为 0.01、0.13、-0.29 亿吨。根据生态环境部发布的《全国碳市场发展报告》（2024 年）、《全国碳市场发展报告》（2025），全国碳排放权交易市场 2021、2022、2023 年度的配额净盈余分别为 147 万吨、1298 万吨及-2900 万吨（配额盈余总量-2900 万吨表示配额缺口为 2900 万吨），净盈余配额总量占应清缴配额量分别为 0.03%、0.25%及-0.55%。而对于第二第三履约周期内，生态环境部并未像《全国碳排放权交易市场第一个履约周期报告》（2022 年）正式发布市场上配额缺口企业的年度缺口量，为分别获得配额盈缺数据，假设第二第三履约周期内发电行业配额缺口企业缺口量比例同第一履约周期，则可推算得第二、三履约周期配额缺口企业年度（2021、2022、2023 年度）配额总缺口分别为 1.07 亿吨、1.07 亿吨、1.10 亿吨，进而计算得配额盈余企业年度配额总盈余分别为 1.08 亿吨，1.20 亿吨、0.81 亿吨。

图表2：前三履约周期配额盈缺测算

履约年度	所属履约周期	盈余企业年度配额 总盈余（万 tCO ₂ ）	缺口企业年度配额 总缺口（万 tCO ₂ ）	配额净盈余 （万 tCO ₂ ）	净盈缺率 （%）
2019-2020	一	50500	-18800	31700	3.37%
2021	二	10844	-10697	147	0.03%
2022		11989	-10691	1298	0.25%
2023	三	8112	-11012	-2900	-0.55%

资料来源：《全国碳排放权交易市场第一个履约周期报告》（2022 年）、《全国碳市场发展报告》（2024 年）、《全国碳市场发展报告》（2025）、中信期货研究所

2.2 第四履约周期碳配额盈缺测算

第四履约周期配额盈缺测算不需考虑三大新增行业的影响。第四履约周期配额发放量及应清缴配额量生态环境部暂未公布，且第四履约周期内新纳入钢铁、水泥、电解铝三大行业，新增重点排放单位 1334 家。根据生态环境部 2025 年 9 月 30 日发布的《2024、2025 年度全国碳排放权交易市场钢铁、水泥、铝冶炼行业配额总量和分配方案（征求意见稿）》，钢铁、水泥、电解铝三大行业配额分配方式与发电行业总体一致，2024、2025 年度配额实行免费分配，基于单位产出碳排放量分配配额。其中 2024 年度为三大行业首个履约年度，三大行业各重点排放单位获得的配额与其经核查的实际碳排放量相等，即无配额缺口，无实质性履约压力。因此，在对第四履约周期配额盈缺进行测算时，暂不考虑新增三大行业的影响。

1. 第四履约周期配额分配总量

根据《2023、2024 年度全国碳排放权交易发电行业配额总量和分配方案》的指导要求，2023、2024 年度配额全部实行免费分配，采用基准法并结合机组层面豁免机制核定机组应发放配额量。将重点排放单位拥有的所有机组对应的年度应发放配额量加总，并结合重点排放单位层面豁免机制得到各重点排放单位年度应发放配额量。配额总量与发电行业控排企业火力发电量呈正比，但因配额分配过程中还需综合考虑履约优惠政策、机组调峰修正系数、配额履约缺口率 20% 上限豁免机制等的影响，以及各机组各自发电量数据统计口径不一，每年度发电行业的配额分配量和应清缴配额总量不能直接计算所得，为简单测算，本报告在生态环境部发布的各年度碳排放平衡值和基准值的基础上，引入综合碳排放平衡值和综合碳排放基准值修正系数的概念，以简化配额分配量及应清缴配额量的计算——根据规模以上火力发电量进行配额分配量及应清缴配额量的计算，计算公式如下：

$$\text{年度应清缴配额量} = \text{年度碳排放强度平衡值} \times \text{综合碳排放平衡值修正系数} \alpha_1 \times \text{年度火力发电量}$$

$$\text{年度碳排放配额分配量} = \text{年度碳排放强度基准值} \times \text{综合碳排放基准值修正系数} \alpha_2 \times \text{年度火力发电量}$$

综合碳排放平衡值修正系数 α_1 ——综合考虑 300MW 等级以上常规燃煤机组、300MW 等级及以下常规燃煤机组、燃煤矸石煤泥水煤浆等非常规燃煤机组（含燃煤循环流化床机组）、燃气机组的发电量占

比，对各机组碳排放平衡值进行加权修正后，用于直接计算应清缴碳配额量的系数；

综合碳排放基准值修正系数 α_2 ——综合考虑 300MW 等级以上常规燃煤机组、300MW 等级及以下常规燃煤机组、燃煤矸石煤泥水煤浆等非常规燃煤机组（含燃煤循环流化床机组）、燃气机组的发电量占比，对各机组碳排放基准值进行加权修正后，用于直接计算配额分配量的系数；

根据 2023 年度已公布配额分配量及应清缴配额量，计算得综合碳排放平衡值和综合碳排放基准值修正系数为 1.0344、1.0355；

《2023、2024 年度全国碳排放权交易发电行业配额总量和分配方案》中给出了 2023 年碳排放平衡值和基准值以及 2024 年碳排放基准值，见下表：2024 年碳排放基准值较 2023 年下降 0.5%。

图表3：2023、2024 各机组发电基准值

机组类别	发电基准值 tCO ₂ /MWh			供热基准值 tCO ₂ /GJ		
	2023 年 平衡值	2023 年 基准值	2024 年 基准值	2023 年 平衡值	2023 年 基准值	2024 年 基准值
300MW 等级以上常规燃煤机组	0.7982	0.7950	0.7910	0.1041	0.1038	0.1033
300MW 等级以下常规燃煤机组	0.8155	0.8090	0.8049			
燃煤矸石、煤泥、水煤浆等非常规燃煤机组 (含燃煤循环流化床机组)	0.8352	0.8285	0.8244			
燃气机组	0.3239	0.3305	0.3288	0.0525	0.0536	0.0533

资料来源：《2023、2024 年度全国碳排放权交易发电行业配额总量和分配方案》、中信期货研究所

假设 2024 年度综合碳排放平衡值和综合碳排放基准值修正系数与 2023 年度相等，则根据 2024 年度规模以上火力发电量，计算得第四履约周期配额分配量及应清缴配额量为：

a)假设一：2024 年碳排放平衡值与 2023 年相等，则第四履约周期配额分配量为 52.79 亿吨，应清缴配额量为 53.35 亿吨，年度配额净盈余为-5603 万吨；

b)假设二：2024 年碳排放平衡值较 2023 年下降 0.5%，则第四履约周期配额分配量为 52.79 亿吨，应清缴配额量为 53.08 亿吨，年度配额净盈余为-2935 万吨；

考虑到中国自双碳目标提出以来，火力发电行业碳排放强度逐年下降，故在计算第四履约周期内电力行业应清缴配额量时，宜适当下调碳排放强度平衡值，本报告采用假设二下的测算结果，即第四履约周期电力行业配额分配量为 52.79 亿吨，应清缴配额量为 53.08 亿吨，年度配额净盈余为-2935 万吨。

2. 第四履约周期配额盈缺测算

为测算第四履约周期配额盈余企业总盈余和配额缺口企业总缺口，假设第四履约周期内发电行业配额缺口企业缺口率同前三履约周期相同，即占应清缴配额总量的 2.1%，则在假设二下，第四履约周期配额缺口企业年度总缺口为 1.11 亿吨，配额盈余企业年度总盈余为 0.82 亿吨。

3. 配额结转政策影响

3.1 配额结转政策要求

重点排放单位碳排放配额最大可结转量计算公式：

$$T = B + NS \times R \quad (\text{公式 1})$$

式中：

T—重点排放单位最大可结转量，单位：tCO₂；

B—重点排放单位基础可结转量，单位：tCO₂；

NS—净卖出配额量，等于买入配额量-卖出配额量，单位：tCO₂；

R—结转倍率，结转倍率设为 1.5；

其他要求：

(1) 可结转量不高于重点排放单位 2025 年期末持仓量；重点排放单位基础可结转量为 10000 吨；净卖出量计算时间范围：2024 年 1 月 1 日至 2025 年 12 月 31 日，如计算结果小于 0 则取值为 0；重点排放单位最大可结转量计算结果向下取整；期末持仓量为 2025 年 12 月 31 日 23:59 全国碳排放权注册登记系统显示的 2019-2024 年配额持仓量；

(2) 全国碳排放权注册登记机构将分 2 次进行配额回收与结转工作：2026 年 3 月 27 日前处理截至 2026 年 3 月 20 日提交的配额结转申请；6 月 26 日前处理 2026 年 3 月 21 日至 2026 年 6 月 10 日提交的配额结转申请；

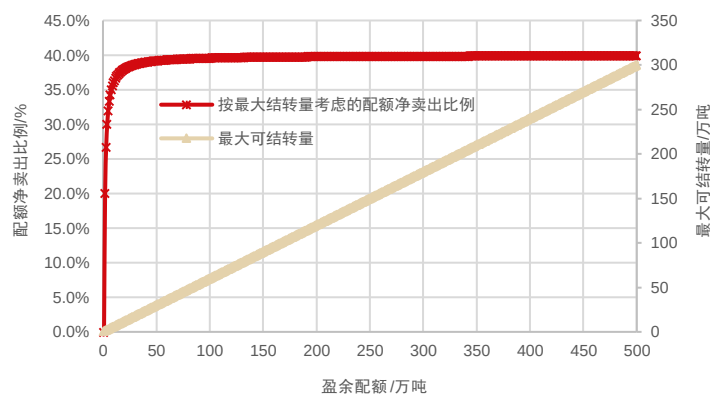
根据公式 1 可知，重点排放单位的最大可结转量与企业 2024 年 1 月 1 日-2025 年 12 月 31 日期间的净卖出量呈正比，且所有企业均享有 1 万吨的基础可结转量。

3.2 最大结转交易策略

(1) **配额缺口企业**：大部分配额缺口企业碳配额账户内往往无多余碳配额，须从市场上购买配额以保证企业足额履约。此类企业净卖出量通常为负，因此最大可结转量为 1 万吨；

(2) **配额盈余企业**：配额盈余企业需根据公式 1 确定其账户内的盈余配额(扣除当年应清缴配额量)，盈余配额需部分卖出，剩余配额结转，否则在结转政策给定的交易截止日期之后，企业碳配额账户内超出最大可结转量部分的配额将被作废，因此配额盈余企业需规划好账户内的结转配额量与为获得配额可结转量而必须被动卖出的配额量(下称“被动卖出配额量”)。

若配额盈余企业“惜售”，则企业或倾向于采用盈余配额尽可能多结转的“保证最大可结转量的交易策略”(下称“最大结转交易策略”)，即企业在此策略下被动卖出配额，并计划将未出售的剩余配额全部结转，以应对未来年度的履约需求，则对采用此交易策略的企业而言，其被动卖出量与被动卖出比例与账户配额量的关系如下图所示。

图表4：被动卖出量与被动卖出比例与账户配额量关系

资料来源：中信期货研究所

从图表 4 中可知，若企业采用“最大结转交易策略”，则对于配额盈余较多的企业，如盈余配额超过 100 万吨，则企业可结转配额量与被动卖出配额量则分别为 60.4 万吨和 39.6 万吨，即要求企业 2024 年 1 月 1 日-2025 年 12 月 31 日期间卖出其盈余配额的约 40%，以保证剩余配额可以在 2026 年顺利完成结转。对于配额盈余数量较少的企业，其被动卖出配额的比例低于 40%，如配额盈余量为 10 万吨、5 万吨、1 万吨的企业，其被动卖出配额的比例分别为 36.0%、32.0%和 0.0%。综上讨论可知，若所有配额盈余企业均采用“最大结转交易策略”，则所有企业被动卖出的配额比例占市场上配额盈余企业总盈余的比例小于 40%。

然而也需注意到，不同企业对碳配额价格走势、全国碳市场政策研判不同以及企业过往交易参与度不同，也会直接或间接影响企业采用不同的配额交易策略，配额盈余企业并不全部选择“最大可结转量交易策略”，或会“主动超额卖出”自己的盈余配额，即配额盈余企业卖出比例超过 40%，甚至全部卖出。在此情况下，配额盈余企业“主动”和“被动”卖出的配额总量或超过 40%。

4. 全国碳市场展望

4.1 被动卖出的配额量测算

为测算市场被动卖出配额量余额，需构建市场配额平衡表。

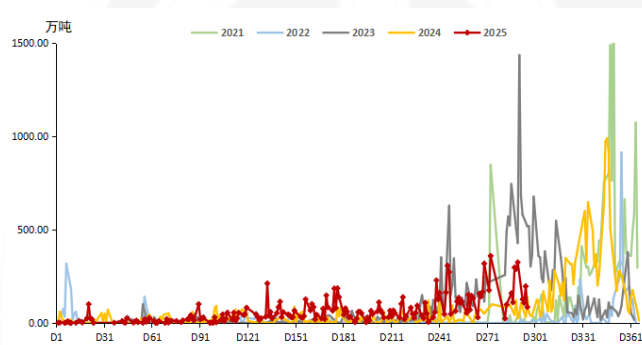
2024 年初全国碳市场碳配额净盈余约 3.68 亿吨。假设 2023 年底，第一履约周期未完全履约的企业通过企业自购补足、主管部门代扣代缴等方式实现 100%履约，则根据前文中对第一第二履约周期配额盈缺的测算，同时考虑第一第二履约周期内 CCER 的使用以抵消部分配额，可测算得 2024 年初全国碳市场净盈余约 3.68 亿吨，其中配额盈余企业总盈余约 3.94 亿吨，配额缺口企业缺口为 0.26 亿吨（第二履约周期未完成履约的企业产生的配额缺口）。

2024 年全年，全国碳市场碳配额年度缺口约 0.29 亿吨。2024 年新分配的年度配额小于重点排放单位应清缴配额量，年度配额的发放首次出现缺口，缺口约 2900 万吨，根据表 2 中的测算，2024 年配额盈余企业新增盈余约 0.81 亿吨，配额缺口企业新增缺口约 1.10 亿吨。

2025 年初，全国碳市场碳配额净盈余约 3.45 亿吨。假设 2024 年底，第二履约周期中未完成履约的企业完成 100% 履约，综合考虑第三履约周期内 CCER 的使用以抵消部分配额，可测算得 2025 年初全国碳市场配额净盈余约 3.45 亿吨，其中配额盈余企业总盈余约 3.46 亿吨，配额缺口企业缺口为 0.01 亿吨（第三履约周期未完成履约的企业产生的配额缺口）。

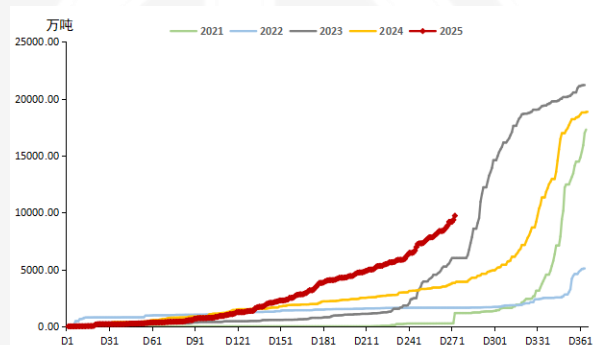
2025 年全年，全国碳市场碳配额年度缺口约 0.29 亿吨（第四履约周期电力行业的配额于 2025 年 8 月 20 日前完成分配发放工作）。第四履约周期新增钢铁、水泥、电解铝三大行业，然而根据《2024、2025 年度全国碳排放权交易市场钢铁、水泥、铝冶炼行业配额总量和分配方案》（征求意见稿），钢铁、水泥、电解铝三大行业 2024 年度各重点排放单位获得的配额与其经核查的实际碳排放量相等，即三大行业配额净盈余量为 0，对全国碳市场配额总盈余无影响。然而若新增的 1334 家控排企业若按基础结转量进场采购碳配额并在 2026 年申请结转，将会给市场带来 1334 万吨的采购需求。

图表5：2021-2025 CEA 日成交量对比



资料来源：Wind 中信期货研究所

图表6：2021-2025 CEA 累计成交量对比



资料来源：Wind 中信期货研究所

截止 2025 年 9 月 30 日，全国碳市场配额成交量为 9809.88 万吨。为计算净卖出量，需确定全国碳市场碳配额成交量中缺口企业的采购需求占比。当年度缺口企业用于履约实际采购的配额量等于当年度配额缺口+上一年度未履约量-CCER 抵消量，计算得 2024 及 2023 年缺口企业用于履约采购的配额占年度配额交易量的比重分别为 68.2%和 70.1%。若假设 2025 年市场大部分参与者的交易行为并没有发生较大变化，且四个季度的配额交易中由履约需求主导的配额交易占比保持一致，则 2025 年前三个季度，由履约采购需求主导的配额交易量约为 6866.92 万吨，约占前三个季度配额交易总量的 70%（交易模型假设一）。然而 2025 交易年度是配额结转政策的截止年，且 2025 年前三季度配额成交量同比增长了 150.3%，若考虑该部分增长均由配额缺口企业采购引起（配额盈余企业为完成结转，此时需卖出配额），则 2025 年前三季度，由履约采购需求主导的配额交易量约为 8633.96 万吨，约占前三个季度配额交易总量的 88%（交易模型假设二）。按过往配额交易规律，配额履约采购需求的释放主要集中在每年的三、四季度，在假设二的前提下，考虑实际交易情况、电力企业排放核算报告及配额预分配时间，第一季度的配额交易中履约需求占比较低，经调整之后，2025 年前三季度由履约采购需求主导的配额交易量约为 8181.50 万吨，约占前三个季度配额交易总量的 83%（交易模型假设三）。

分别计算三种交易模型假设下，2025 年四季度市场被动卖出的配额量：

测算 2025 年全年由履约采购需求主导的配额交易量为时，假设所有重点排放单位均按要求于 2025 年 12 月 31 日前完成履约工作，即履约率 100%；则 2025 全年，缺口企业履约缺口约 1.12 亿吨。

(1) 交易模型假设一：由履约采购需求主导的配额交易量为 6866.92 万吨，约占前三个季度配额交易总量的 70%。在此假设下，综合考虑 CCER 的抵消量、履约完成情况等因素，假设各盈余企业采用保障最大可结转量的“惜售”交易策略，则截止 9 月 30 日，市场被动卖出配额量余额约 2365.62 万吨，采购需求余额约 4385.08 万吨。

(2) 交易模型假设二：由履约采购需求主导的配额交易量约为 8633.96 万吨，约占前三个季度配额交易总量的 88%。在此假设下，综合考虑 CCER 的抵消量、履约完成情况等因素，假设各盈余企业采用保障最大可结转量的“惜售”交易策略，则截止 9 月 30 日，市场被动卖出配额量余额约 599.84 万吨，采购需求余额约 2619.31 万吨。

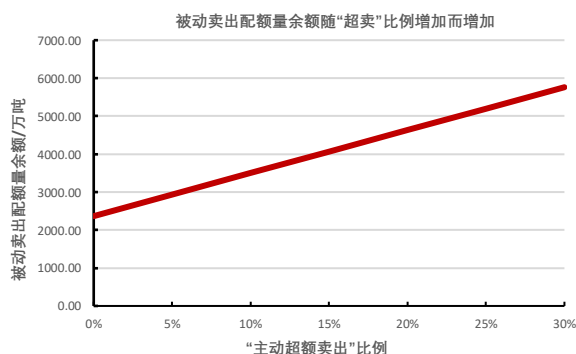
(3) 交易模型假设三：由履约采购需求主导的配额交易量约为 8181.50 万吨，约占前三个季度配额交易总量的 83%。在此假设下，综合考虑 CCER 的抵消量、履约完成情况等因素，假设各盈余企业采用保障最大可结转量的“惜售”交易策略，则截止 9 月 30 日，市场被动卖出配额量余额约 1090.33 万吨，采购需求余额约 3109.80 万吨。

4.2 主动超额卖出量的影响

正如上文中提到的配额盈余企业并不全部选择“最大可结转量交易策略”，或会“主动超额卖出”自己的盈余配额，即盈余配额卖出比例超过 40%，甚至全部卖出。在此情况下，配额盈余企业“主动”和“被动”卖出的配额总量将超过 40%。配额盈余企业“主动超额”卖出配额的比例越大，则四季度被动卖出配额量的余额越多，市场抛售压力越大。三种交易模型假设下，四季度市场被动卖出配额量的余额随配额盈余企业主动超额卖出的比例变化如下列图表所示。

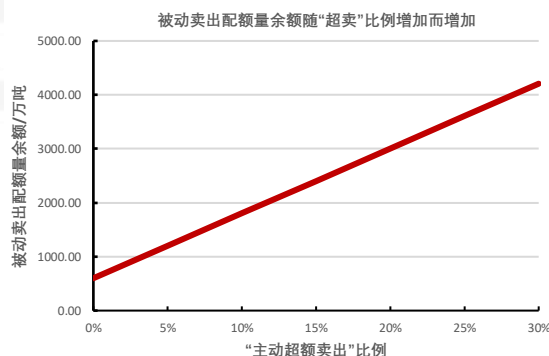
考虑到三种假设的局限性，图表 10 进一步给出了交易模型假设一情况下，市场四季度被动卖出配额余额与履约采购余额随 2025 年前三季度全国碳市场中履约交易占比变化的走势：履约交易占比越高，则剩余的被动卖出配额量越少，则剩余的履约采购量越少。

图表7：被动卖出配额量余额与“主动超额卖出”比例之间的关系（假设一）



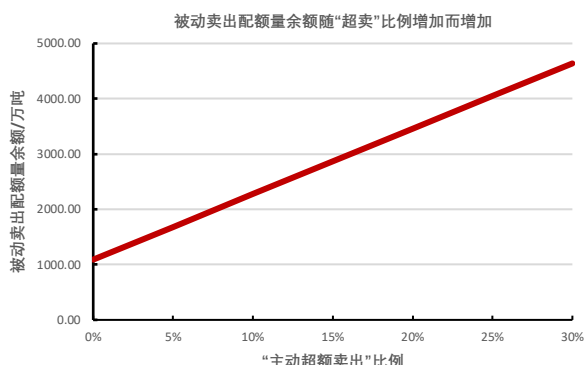
资料来源：中信期货研究所

图表8：被动卖出配额量余额与“主动超额卖出”比例之间的关系（假设二）



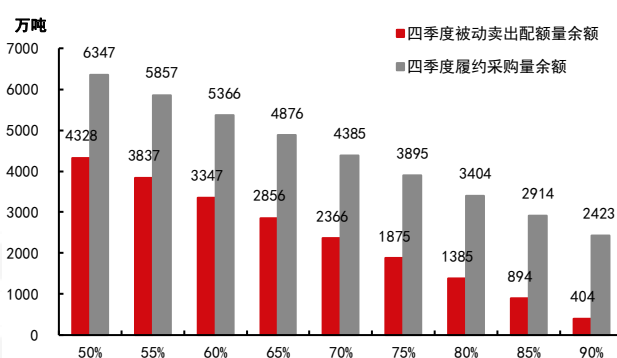
资料来源：中信期货研究所

图表9：被动卖出配额量余额与“主动超额卖出”比例之间的关系（假设三）



资料来源：中信期货研究所

图表10：被动卖出配额余额及履约采购余额（假设一）



资料来源：中信期货研究所

4.3 碳价或震荡偏强

企业出于“惜售”的考虑，配额卖出意愿降低。10月全国碳市场成交量为4156.25万吨，较9月增长了27.1%，被动卖出的配额余额进一步减小。经测算，在三种交易模型假设下，若不考虑企业的主动超额卖出情况，被动卖出的配额余额已等于0，表明配额盈余企业剩余配额或已满足最大可结转量的要求，企业出于“惜售”的考虑，配额卖出意愿降低。随着配额缺口企业采购的进行，配额履约需求量也进一步减少（仍有部分企业未完成配额的履约采购，即需求量>0）。且除配额履约需求之外，或有企业考虑利用企业的基础结转量政策，在碳价较低时采购配额并进行结转（企业认为当前碳价被低估时，或会采用此策略），该部分潜在的采购需求约2000万吨左右。

因此，碳价底部支撑或已得到确认，随着11月履约需求的进一步释放，碳价或将持续反弹至履约需求退场（12月上旬左右）。在此期间，碳价快速上涨过程中或有回调，但短期调整空间有限。需持续关注配额日成交量情况，若连续缩量则表示履约需求已释放完毕，配额交易进入淡季，价格或会有回调风险。

长期来看，预期2026年碳市场重回供应偏紧状态，碳价或震荡上行。供应端，2026年初全国碳市场配额净盈余将进一步减少，配额交易或再次转向“惜售”逻辑，市场重回供应偏紧的状态。需求端，2026年度，新增的三大行业配额缺口企业将为市场带来新的需求增量。

操作建议：（1）配额缺口企业可逢低买入配额完成年度履约；（2）配额盈余企业需按照结转政策要求确认剩余配额是否已满足最大可结转量要求，超额部分可选择逢高卖出；（3）配额净买入企业及新纳入全国碳市场的三大行业的重点排放单位享有1万吨基础结转量，可关注碳市场逢低买入机会。

但仍需注意：重点排放单位配额交易策略的多样性或会使三种交易模型假设情况偏离真实交易情况；此外市场履约需求占配额交易量的比重及其在各个月份的分布情况均会影响到本篇报告对企业净卖出量测算的准确性；需求端潜在需求的实际转化情况不能确定；少部分企业或因决策失误等原因导致12月下旬疯狂抛售多余配额而造成碳价大幅波动，此为不可预测因素，因此企业应尽量避免在年底碳价可能出现巨幅波动时进行交易。

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